

Removable Boundaries and Rigidity of Circle Domains

OpenAI

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Abstract

We prove that a circle domain with conformally removable boundary is conformally rigid, with no restriction on the number of complementary components. This establishes the removability-to-rigidity direction of the He–Schramm Conjecture.

1 Introduction

A circle domain is a nonempty connected open subset of the Riemann sphere $\widehat{\mathbb{C}} = \mathbb{C} \cup \{\infty\}$ whose complementary components are points or closed round disks; the complement may be empty. Circle boundaries give a particularly concrete conformal model of a plane domain. Their usefulness depends on two separate questions: whether such a model exists, and whether its conformal structure determines its placement in the sphere up to a Möbius transformation. This paper addresses the second question under conformal removability of the boundary.

A compact set $E \subset \widehat{\mathbb{C}}$ is *conformally removable* if every orientation-preserving homeomorphism of the sphere that is conformal on $\widehat{\mathbb{C}} \setminus E$ is Möbius. A circle domain is *conformally rigid* if every conformal equivalence from it onto another circle domain is the restriction of a Möbius transformation. The definitions have different starting data: removability concerns a map already defined on the sphere, whereas rigidity concerns only a map between the domains.

Theorem 1.1. *Every circle domain with conformally removable boundary is conformally rigid. More explicitly, if $\Omega, \Omega' \subset \widehat{\mathbb{C}}$ are circle domains, $\partial\Omega$ is conformally removable, and $f: \Omega \rightarrow \Omega'$ is a conformal equivalence, then there is a Möbius transformation M such that $f = M|_{\Omega}$.*

The theorem allows arbitrary cardinality of the complementary component set. The required boundary control is a conclusion of the argument; no boundary extension of f is part of the hypothesis.

1.1 Rigidity and removability

He and Schramm proved circle-domain uniformization for countably connected domains and rigidity for countably connected circle domains [6]. Their subsequent rigidity theorem permits uncountably many complementary components when the boundary has σ -finite linear measure [7]. They conjectured that a circle domain is rigid if and only if its boundary is conformally removable. The forward implication in this proposed equivalence is false: Rajala constructed a rigid circle domain with a non-removable boundary [15, Theorem 1.1]. Theorem 1.1 establishes the remaining removability-to-rigidity implication positively.

Several intermediate results clarify the role of boundary geometry. Younsi studied the formulations involving removability of the whole boundary and of its Cantor subsets, and proved that

conformal rigidity is equivalent to quasiconformal rigidity [17, Theorems 3–5]. Ntalampekos and Younsi proved rigidity under square integrability of the quasihyperbolic distance in a bounded portion of the domain; this includes Hölder and John circle domains [13, Theorem 1.6 and Corollary 1.7]. Ntalampekos subsequently proved rigidity when every compact subset of the set of point boundary components is *countably negligible for extremal distance*, or CNED [11, Theorem 1.2]. For every pair of disjoint nondegenerate continua, the CNED condition says that restricting the joining curves to those that meet the exceptional set in at most countably many points leaves their conformal modulus unchanged. These results develop methods for controlling boundary traces under explicit geometric or extremal-distance hypotheses.

Continuous Sobolev extension is a second relevant line of work. Jones and Smirnov related removability for continuous Sobolev functions to quasiconformal removability and proved criteria based on the geometry near a domain boundary [8, Definition 2 and Theorems 2, 4]. For closed CNED sets, Ntalampekos proved continuous $W^{1,n}$ extension in dimension n [12, Theorem 1.4]. Our intermediate Theorem 5.1 treats compact totally disconnected planar sets under conformal removability itself: a scalar function continuous across such a set, with finite Dirichlet energy on its complement, is globally Sobolev. Continuity across the set is an essential hypothesis of this intermediate statement.

1.2 The proof and its quantitative input

The first difficulty is to use removability before a conformal map is known to extend to the sphere. We handle it initially for the complement of a compact totally disconnected set B . A normalized conformal map h of $\widehat{\mathbb{C}} \setminus B$ may replace a point of B by a nonpoint complementary component in its image. Component correspondence supplies a continuous inverse collapse map π_h on the target sphere. For a continuous scalar function u on the source sphere, the function $u \circ \pi_h$ therefore has the prescribed constant value on each target component. We call h a model retaining u when this continuation is globally Sobolev and its gradient vanishes on the omitted set.

The companion paper [14] proves a quantitative retained-test transfer and selection theorem. We use its scalar form in Section 3: compatible finite lists of scalar functions are retained exactly on exhausting cores, while the remaining energy and the square sum of boundary oscillations tend to zero. Compatibility here is the condition that the joint image of B under each pair of tests has planar area zero. Filling the finite round holes and passing to a weak limit produces the required models. A separate contour argument identifies their values on all omitted components, including point components whose union can have positive area. This is the quantitative part of the uniformization theorem needed in this paper.

The next step varies the model. Among maps retaining a fixed finite list, we nearly maximize the real part of the first Laurent coefficient at infinity. A conductivity deformation, constructed through the classical Beltrami equation, then gives a scalar height close to the ordinary vertical coordinate, with arbitrarily small Dirichlet error. A level-set argument makes this height compatible with the retained tests. After flattening the countably many values belonging to nonpoint holes, a horizontal coordinate can be added as well. Together the two coordinates yield barriers around all source ends with arbitrarily small energy.

We save these barriers while enlarging the finite test lists. If a limiting model had a nonpoint omitted component, its projection on one axis would contain an interval of positive length. Every transverse line over that interval would force a saved barrier to change from one to zero in a bounded distance. The resulting fixed lower bound on its energy contradicts the construction. Thus every omitted component is a point, and component correspondence extends the limiting

model to a sphere homeomorphism. Removability makes the normalized limit the identity; weak Sobolev compactness then proves the scalar extension theorem.

Finally we return to a conformal equivalence $f : \Omega \rightarrow \Omega'$. Point components may correspond to disks, and disks to points, so direct continuation of a coordinate of f need not be continuous. We mask the countably many disk-containing pairs and apply the scalar theorem to the remaining common point components. The masks show that their traces have null axial projections on almost every line. This use of linewise variation is related to the Sobolev-removability arguments of Jones and Smirnov [8] and to the boundary-control methods in [13, 11]; the mask construction here accommodates all common point components simultaneously. Integrating the resulting slice estimates gives a flux inequality. Comparison with source and target area bounds forces equality, and hence forces f to be Möbius.

Section 2 constructs component correspondence and records normalization bounds. Section 3 obtains models retaining prescribed tests. Section 4 supplies the conductivity and level-set estimates, and Section 5 uses them to prove scalar Sobolev extension. Section 6 constructs the masks and proves the trace statement. Section 7 completes the flux and area argument.

2 Contours, ends, and normalization

Write $D_R = \{z \in \mathbb{C} : |z| < R\}$. Unless specified otherwise, gradients and area elements in a Dirichlet integral are Euclidean. In dimension two that integral is conformally invariant; global $W^{1,2}(\widehat{\mathbb{C}})$ uses spherical area for the function norm. A *null family* of compact sets has only finitely many members of diameter greater than any given positive number.

The following topological fact concerns components, rather than a pointwise extension across their boundaries.

Lemma 2.1 (Component correspondence). *Let $g : V \rightarrow V'$ be an orientation-preserving homeomorphism of sphere domains, with $\infty \in V \cap V'$ and $g(\infty) = \infty$. There is a bijection $K \mapsto K'$ between their complementary components. The relation*

$$\mathcal{R}_g(z) = \begin{cases} \{g(z)\}, & z \in V, \\ K', & z \in K \subset \widehat{\mathbb{C}} \setminus V \end{cases}$$

is upper semicontinuous: for every complementary component K and every open neighborhood W of K' , some open neighborhood O of K satisfies $\mathcal{R}_g(z) \subset W$ for all $z \in O$. In particular the relation has closed graph. The analogous assertion holds for g^{-1} .

If $\widehat{\mathbb{C}} \setminus V = B$ is totally disconnected, the inverse map g^{-1} extends continuously to the sphere by collapsing each target component to its corresponding point of B . If both complements are totally disconnected, g extends to an orientation-preserving homeomorphism of the sphere.

Proof. We give the contour construction from the component quotient, using Moore's classical decomposition theorem. Collapse every component of $F = \widehat{\mathbb{C}} \setminus V$ to a point, leaving domain points as singletons. This decomposition has closed equivalence relation. For convergent related pairs, a subsequence in domain singleton classes forces equal limits. Otherwise pass to a Hausdorff limit of complementary components; it is a continuum in F containing both limiting points, which therefore remain in the same component. Every component K is nonseparating. Indeed V lies in one component of $\widehat{\mathbb{C}} \setminus K$; any other component A would miss V , making $K \cup \overline{A}$ a connected subset of F strictly larger than K . Moore's Decomposition Theorem [10] therefore gives a closed monotone quotient $\pi : \widehat{\mathbb{C}} \rightarrow X$ onto a topological sphere. Its restriction to V is a homeomorphism

onto an open subset. The omitted set $B_X \subset X$ is compact and totally disconnected: the inverse image of a continuum under a closed monotone map is connected, so a continuum in B_X lifts into a single component of F .

For a physical Jordan curve $C \subset V$ avoiding infinity, $\pi|_C$ is a homeomorphism onto a Jordan curve. Every other decomposition element is connected and misses C , so each physical Jordan side is saturated. Their images are therefore disjoint connected open sets partitioning $X \setminus \pi(C)$, hence are its two Jordan sides; the bounded physical side maps to the side missing $\pi(\infty)$. The quotient disks below are these images of bounded physical disks.

The compact metric space B_X has a clopen base. Partition it into finitely many clopen pieces in small quotient Jordan neighborhoods. Their physical inverse images have disjoint open neighborhoods avoiding the other pieces. Regular levels of smooth cutoffs around these compact inverse images give finitely many smooth curves in V . Fill the outer curves, discard nested ones, and discard any resulting disk that misses the omitted set. Filling stays inside the chosen quotient Jordan neighborhood: the inverse image of its complement is connected, contains infinity, and misses the curve. It therefore lies on the unbounded side. This constructs finite families of physical Jordan disks with disjoint closures and smooth boundaries in V , whose quotient images cover B_X by their interiors.

Refine these covers with new closures inside old disks and quotient mesh tending to zero. Their closed exteriors are connected smooth cores $L_n \subset V$ with

$$L_n \subset \text{int } L_{n+1}, \quad \bigcup_n L_n = V.$$

For $b \in B_X$, let $U_n(b) \subset X$ be its containing quotient disk and let $C_n(b) = \pi^{-1}(\partial U_n(b)) \subset V$ be the corresponding physical boundary. Then

$$\overline{U_{n+1}(b)} \subset U_n(b), \quad \bigcap_n \overline{U_n(b)} = \{b\}.$$

The corresponding physical disks shrink to $\pi^{-1}(b)$; their physical diameters need not tend to zero.

All these curves and cores lie in V , where g is already defined. The compact embedded surface $g(L_n)$ is the closed exterior of the disjoint bounded Jordan disks enclosed by its boundary curves. Orientation fixes their sides, since the interior of the core is connected and contains infinity. Let $Q_n(b)$ be the closed bounded disk enclosed by $g(C_n(b))$. Refinement gives

$$Q_{n+1}(b) \subset \text{int } Q_n(b), \quad K'_b = \bigcap_n Q_n(b).$$

This intersection is a nonempty continuum missing V' , because the image cores exhaust V' . Conversely every point outside V' lies in one image disk at each level. The corresponding nested source disks determine a unique b by their quotient mesh. A connected subset of the target complement cannot cross an image contour, so K'_b is exactly a complementary component. This proves the bijection without any countability assumption.

Given an open $W \supset K'_b$, nested compactness gives $Q_n(b) \subset W$ for sufficiently large n . The physical inverse image of $U_n(b)$ is an open neighborhood of $\pi^{-1}(b)$, and every fiber over this neighborhood lies in $Q_n(b)$. This proves upper semicontinuity. Applying the same argument to g^{-1} proves the inverse assertion. Singleton source fibers then give the stated continuous collapse map. When both sets of fibers are singletons, the extension is a continuous bijection of compact spheres and hence a homeomorphism. Its orientation is fixed by its restriction to V . $\square$

Lemma 2.2 (Normalized maps). *Let $G \subset \widehat{\mathbb{C}}$ be a fixed domain whose complement is contained in D_{R_0} . The univalent conformal maps on G with expansion*

$$h(z) = z + \frac{b(h)}{z} + O(z^{-2}) \quad (z \rightarrow \infty)$$

have uniformly bounded $b(h)$ and uniformly bounded omitted sets. They form a normal family on compact subsets of G , and every convergent subsequence has a nonconstant univalent limit with the same normalization. Uniformly over this family, $h(z) = z + O(|z|^{-1})$ for large $|z|$.

Proof. Apply the Exterior Area Theorem [2] to $h(R_0\zeta)/R_0$. For $h(z) = z + \sum_{n \geq 1} \alpha_n z^{-n}$ it gives

$$\sum_{n \geq 1} n |\alpha_n|^2 R_0^{-2n} \leq R_0^2.$$

Thus $|b(h)| \leq R_0^2$, and Cauchy–Schwarz gives the asserted uniform exterior estimate. In particular $h(\partial D_{2R_0})$ lies in a fixed disk. Its bounded side contains the whole omitted set and $h(G \cap D_{2R_0})$: injectivity prevents an interior image point from crossing this image circle, and points just inside the circle lie on its bounded side. Thus the maps are uniformly bounded on $G \cap D_{2R_0}$ as well as on exterior compact sets, proving local normality. The uniform exterior estimate prevents a constant limit and preserves the normalization. Hurwitz’s Theorem then makes every limit univalent. $\square$

Lemma 2.3 (Area zero). *A conformally removable compact set has spherical area zero. Every compact subset of a conformally removable compact set is conformally removable.*

Proof. If a removable compact set had positive area, choose a bounded measurable subset A of positive planar area in a coordinate chart and a coefficient $\mu = k \mathbf{1}_A$ with $0 < k < 1$. The Measurable Riemann Mapping Theorem [3] gives an orientation-preserving quasiconformal sphere homeomorphism with this Beltrami coefficient. It is conformal off the compact set and is not Möbius, since its coefficient is nonzero on a set of positive area. This is a contradiction. For the second assertion, a homeomorphism conformal off a smaller compact set is also conformal off the larger one, so the defining removability implication applies. $\square$

3 Models retaining finitely many scalar tests

Let $B \subset \mathbb{C}$ be compact and totally disconnected, and put $G = \widehat{\mathbb{C}} \setminus B$. For a normalized univalent map h on G , Lemma 2.1 gives a continuous collapse map $\pi_h : \widehat{\mathbb{C}} \rightarrow \widehat{\mathbb{C}}$: it equals h^{-1} on $h(G)$ and sends the complementary component corresponding to $b \in B$ to b .

Definition 3.1. Let $u_1, \dots, u_m$ be bounded continuous real functions on $\widehat{\mathbb{C}}$ whose restrictions to G belong to $W^{1,2}(G)$. A *model* for this list is a univalent conformal map

$$h(z) = z + \frac{b(h)}{z} + O(z^{-2}) \quad (z \rightarrow \infty)$$

such that its nonpoint complementary components form a finite or countable null family of closed Jordan disks, and

$$U_i = u_i \circ \pi_h \in W^{1,2}(\widehat{\mathbb{C}}), \quad \nabla U_i = 0 \quad \text{a.e. on } \widehat{\mathbb{C}} \setminus h(G).$$

A family is null if only finitely many members have diameter greater than any prescribed positive number.

The conductivity argument below postcomposes models with quasiconformal homeomorphisms. These preserve Jordan disks and null families but need not preserve circles, which is why the model class allows Jordan disks. The existence result below will nevertheless supply round disks.

We require the pairwise trace condition

$$\text{area}((u_i, u_j)(B)) = 0 \quad (i \neq j). \quad (3.1)$$

Only the values on B occur in this condition.

Proposition 3.2. *Every finite list in Definition 3.1 satisfying (3.1) has a model. A model can, in fact, be chosen with circle-domain image.*

The uniformization input below is the finite-transfer construction of [14]. Its control of the retained scalar tests is essential; circle-domain existence by itself would not give Proposition 3.2.

3.1 The precise retained-test input

We use the following specialization of [14, Corollary 7.6]. It is a consequence of the companion's Theorem 7.5, whose selection begins with the prescribed tests and preserves them through all later compatible extensions.

First suppose the tests $u_1, \dots, u_m$ in Proposition 3.2 are smooth on G . For any sequence $\varepsilon_n > 0$ tending to zero, there are nested compact smooth cores $K_n \subset G$, finite bordered completions Q_n containing neighborhoods of K_n with their original conformal structure, normalized conformal maps $f_n : Q_n^\circ \rightarrow \Omega_n$ onto finite circle domains, and smooth transferred tests $v_{i,n}$ on Ω_n with the following properties. Every compact subset of G is contained in $\text{int } K_n$ for all sufficiently large n , and

$$v_{i,n} \circ f_n = u_i \quad \text{near } K_n, \quad (3.2)$$

$$\int_{\Omega_n \setminus f_n(K_n)} |\nabla v_{i,n}|^2 \leq \varepsilon_n^2, \quad (3.3)$$

$$\sum_T A_{i,n}(T)^2 \leq \varepsilon_n^2. \quad (3.4)$$

Here T ranges over the complementary round disks of Ω_n , and the cluster values of $v_{i,n}$ at T lie in an interval of length $A_{i,n}(T)$. The first equality also preserves energy near K_n . The companion actually bounds the sum of the tail energies over the whole retained finite list by ε_n^2 .

These assertions hold on one shared sequence. In particular, every fixed source contour and its collar are retained eventually. If C is an oriented source Jordan contour and $z \in K_n$ lies on its bounded side, then $f_n(z)$ lies on the bounded side of $f_n(C)$; the converse holds for retained-core points. After passage to a subsequence,

$$f_n \longrightarrow h \quad \text{locally uniformly on } G, \quad h(G) \text{ is a circle domain.} \quad (3.5)$$

All omitted disks of the finite domains lie in one fixed disk D_L . For every fixed source dust contour, its bounded target sides also lie in a fixed disk. The omitted-set bound depends only on a fixed exterior radius for G ; the contour bound may also depend on that fixed contour. Neither depends on the number of retained tests.

Here the quotient sphere in the scalar corollary is the source sphere itself, because every source complementary component is a point. Thus the prescribed u_i are continuous on that quotient, smooth on G , and of finite Dirichlet energy there. Their boundary value at b is the

single number $u_i(b)$, and condition (3.1) is precisely the corollary's pairwise trace hypothesis. The corollary retains all these original tests while adding auxiliary tests for the circle-domain selection. Consequently the core agreement, energy and oscillation bounds, and contour assertions above hold simultaneously for the prescribed list. No countability assumption on B is needed.

It remains to turn these finite transferred tests into their prescribed continuous Sobolev continuations on the limiting sphere. That passage, including the values on positive-area point dust, is proved next.

3.2 Filling holes and identifying the limiting tests

We record explicitly the small error introduced by filling round holes. Suppose v is bounded and locally Sobolev on a finite circle domain containing infinity, has finite energy, and has cluster interval of length at most $A(T)$ at each hole T . Fill T by a constant in its cluster interval, call the resulting function $\tilde{v}$, and extend ∇v by zero on the holes to a vector field g . If $r(T)$ is the radius of T , then for $j = 1, 2$ and $\varphi \in C_c^\infty(\mathbb{C})$,

$$\left| \int \tilde{v} \partial_j \varphi + \int g_j \varphi \right| \leq 4 \|\varphi\|_\infty \sum_T r(T) A(T). \quad (3.6)$$

Indeed, on almost every line parallel to the j th axis the function is locally absolutely continuous on each domain interval and has finite variation on its bounded portions relevant to the support of the test function. At either endpoint of a hole the jump to its chosen constant is at most $A(T)$. Integrating the two endpoint errors over the transverse projection of that hole, of length $2r(T)$, gives (3.6). Point holes affect only finitely many transverse parameters. Compactly supported test functions remove any issue at infinity.

Fix one original smooth test $u = u_i$. Clip $v_{i,n}$ to a fixed closed interval containing the range of u . This preserves (3.2), decreases energy and cluster oscillations, and supplies a uniform bound. Fill the holes as above, obtaining $\tilde{v}_n$, and write g_n for the domain gradient extended by zero over the holes. Disjointness and the common target bound imply

$$\sum_T r_n(T)^2 \leq L^2, \quad \sum_T r_n(T) A_{i,n}(T) \leq L \varepsilon_n.$$

Moreover, writing $E_u = \int_G |\nabla u|^2$, we have

$$\int |g_n|^2 \leq E_u + \varepsilon_n^2. \quad (3.7)$$

Take weak subsequential limits of the bounded functions in spherical L^2 and of their differentials in L^2 . Equation (3.6) shows that the limiting differential is the weak derivative of the limiting function V . This argument takes place in the planar chart containing all holes; near infinity the retained core gives the ordinary Sobolev derivatives. Hence $V \in W^{1,2}(\hat{\mathbb{C}})$ and $\int |\nabla V|^2 \leq E_u$. On compact subsets of $h(G)$, inverse convergence and (3.2) give

$$V = u \circ h^{-1} \quad \text{a.e. on } h(G).$$

By conformal invariance the energy of this function on $h(G)$ is already E_u . Thus

$$\nabla V = 0 \quad \text{a.e. on } F := \hat{\mathbb{C}} \setminus h(G). \quad (3.8)$$

Zero gradient alone does not identify values on a positive-area set of point components. We now determine those values. Fix a source dust disk U with smooth boundary $C \subset G$, and an

interval $[\alpha, \beta]$ containing $u(\overline{U})$. Retain C and a collar in every sufficiently late core. Let O_n and O be the bounded sides of $f_n(C)$ and $h(C)$, respectively. Define

$$e_n^+ = \begin{cases} (\tilde{v}_n - \beta)_+ & \text{on } O_n, \\ 0 & \text{off } O_n, \end{cases} \quad e_n^- = \begin{cases} (\alpha - \tilde{v}_n)_+ & \text{on } O_n, \\ 0 & \text{off } O_n. \end{cases}$$

Every retained-core point mapping into O_n comes from U , by the contour-side correspondence. Both excesses therefore vanish on those core pieces and on an inner collar of $f_n(C)$. Extending them by zero creates no jump across that contour. Their extended gradient norms are at most ε_n , by (3.3), and their hole-jump errors satisfy (3.6), since positive-part maps are 1-Lipschitz. They are uniformly bounded and supported in one fixed disk. Every weak limit consequently has zero distributional gradient and bounded support, and is zero.

To pass this conclusion to V , take a nonnegative smooth function φ compactly supported in O . Winding-number stability puts its support in O_n eventually. Therefore

$$\int \varphi(\tilde{v}_n - \beta) \leq \int \varphi e_n^+ \rightarrow 0, \quad \int \varphi(\alpha - \tilde{v}_n) \leq \int \varphi e_n^- \rightarrow 0.$$

Weak convergence gives $\alpha \leq V \leq \beta$ almost everywhere on O . In particular, we have not interchanged a nonlinear operation with a weak limit.

For every integer $\ell \geq 1$, choose one finite source dust-disk cover of B so fine that u oscillates by less than $1/\ell$ on each closed disk. Apply the preceding conclusion to those finitely many intervals. The union of all their exceptional null sets, over all ℓ , is still null. If $w \in F$ is outside this union and its source end is b , the whole component containing w lies inside the image contour of each cover disk containing b . It follows that $|V(w) - u(b)| \leq 1/\ell$ for every ℓ , and hence $V(w) = u(b)$. This also identifies the values on a collection of point components of positive total area.

The continuous function $u \circ \pi_h$ now agrees almost everywhere with V . It is the required Sobolev representative, with the prescribed value on *every* complementary component, and (3.8) gives its zero gradient there. The argument applies to each of the finitely many tests. Finally, the nonpoint components of F are disjoint round disks in a bounded region. Their squared radii have finite sum, so they are countable and form a null family. This proves Proposition 3.2 for smooth tests.

3.3 Exact traces under smoothing

We use the following elementary extension fact.

Lemma 3.3. *Let $F \subset \widehat{\mathbb{C}}$ be closed. If $w \in C(\widehat{\mathbb{C}}, \mathbb{R})$ vanishes on F , is locally Sobolev on $\widehat{\mathbb{C}} \setminus F$, and has finite Dirichlet energy there, then $w \in W^{1,2}(\widehat{\mathbb{C}})$ and $\nabla w = 0$ almost everywhere on F .*

Proof. For $\delta > 0$ the signed truncation $w_\delta = \text{sgn}(w)(|w| - \delta)_+$ has compact support away from F . It is globally Sobolev by localization, and its gradient norm is at most that of w off F ; its gradient is zero on F . Since $\|w_\delta - w\|_\infty \leq \delta$, weak compactness proves the assertion, including the vanishing gradient on F . $\square$

For a continuous test u as in the proposition, choose a locally finite smooth partition of unity (ρ_k) on G , with each support compact in G . Local mollification gives smooth functions a_k near these supports, with both uniform error and local gradient error at most η_k . Choose the positive η_k so that

$$\eta_k \rightarrow 0, \quad \sum_k \eta_k (1 + \|\nabla \rho_k\|_2) < \infty.$$

Then $\hat{u} = \sum_k \rho_k a_k$ is bounded and smooth on G , and the product rule shows that $\nabla \hat{u} - \nabla u \in L^2(G)$. As a point approaches B , every fixed finite collection of supports is eventually avoided. The bound $|\hat{u} - u| \leq \max\{\eta_k : \rho_k \neq 0\}$ therefore tends to zero there. Defining $\hat{u} = u$ on B gives a continuous function on the sphere with exactly the original trace. The same construction in a chart at infinity is included in the partition.

Apply this to each test. Their traces and the pairwise condition (3.1) are unchanged. The smooth case provides one circle-domain model h for the $\hat{u}_i$. Each function

$$w_i = (u_i - \hat{u}_i) \circ \pi_h$$

is continuous, vanishes on F , is locally Sobolev on $h(G)$, and has finite energy there by conformal invariance. Lemma 3.3 makes it globally Sobolev with zero gradient on F . Adding it to $\hat{u}_i \circ \pi_h$ proves the model properties for the original u_i . If B is empty, the identity is already a model. Proposition 3.2 follows in all cases.

4 Almost straight coordinates and null trace pairs

Fix a compact totally disconnected set $B \subset \mathbb{C}$, put $G = \hat{\mathbb{C}} \setminus B$, and fix a finite list of tests satisfying (3.1). Let $\mathcal{M}$ be its nonempty class of models from Definition 3.1. Write

$$h(z) = z + \frac{b(h)}{z} + O(z^{-2}), \quad \beta = \sup_{h \in \mathcal{M}} \Re b(h).$$

Lemma 2.2 makes β finite and places all model omitted sets in one fixed disk. In this section $w = X + iY$ denotes the target coordinate.

We construct a height coordinate close to the usual vertical coordinate on compact sets, with a small global gradient error. The subsequent level-set argument will allow it to be combined with the retained tests while preserving the pairwise trace condition.

Proposition 4.1. *Suppose $d > 0$ and $h \in \mathcal{M}$ satisfies $\Re b(h) > \beta - d$. Set $F = \mathbb{C} \setminus h(G)$. There is a continuous $v \in W_{\text{loc}}^{1,2}(\mathbb{C})$, constant on every component of F , such that $\nabla v = 0$ almost everywhere on F and*

$$\int_{\mathbb{C}} |\nabla(v - Y)|^2 dA \leq 2\pi d. \quad (4.1)$$

The function v is a locally uniform limit of imaginary coordinates of normalized plane homeomorphisms conformal outside a fixed disk. Moreover, for each compact $K \subset \mathbb{C}$, the quantities $\|v - Y\|_{C(K)}$ tend to zero as $d \downarrow 0$, uniformly over these choices of nearly maximizing models. The exterior bounds used here depend only on B .

Proof. For an integer $n \geq 2$, set $k = n$ on F and $k = 1$ elsewhere. We first construct the principal quasiconformal coordinate $\phi_n = a_n + iv_n$ satisfying

$$(a_n)_X = k(v_n)_Y, \quad (a_n)_Y = -k(v_n)_X, \quad \phi_n(w) = w + \frac{c_n}{w} + O(w^{-2}). \quad (4.2)$$

Indeed this system is equivalent to the real-linear Beltrami equation

$$(\phi_n)_{\bar{w}} = q \overline{(\phi_n)_w}, \quad q = \frac{k-1}{k+1}.$$

We use the classical Cauchy–Beurling construction of the Beltrami solution [1, Sections 1–3]. Let $\mathcal{C}$ be the plane Cauchy transform and $\mathcal{S} = \partial\mathcal{C}$ its L^2 -isometric Beurling transform, with $\bar{\partial}\mathcal{C}g = g$. The equation

$$g = q \overline{1 + \mathcal{S}g}$$

is an affine contraction on $L^2(\mathbb{C})$ considered as a real Banach space. Its solution is supported on F . Thus $\phi_n = w + \mathcal{C}g$ is locally Sobolev and has the expansion in (4.2); compact support also makes g integrable. Set $\mu = (\phi_n)_{\bar{w}}/(\phi_n)_w$ where the denominator is nonzero, and $\mu = 0$ otherwise. The distortion inequality $|(\phi_n)_{\bar{w}}| \leq \|g\|_\infty |(\phi_n)_w|$ shows that $(\phi_n)_{\bar{w}} = \mu(\phi_n)_w$ almost everywhere, $\|\mu\|_\infty < 1$, and $\mu = 0$ off F . Standard planar weak-solution regularity, the Measurable Riemann Mapping Theorem, and holomorphic factorization [1, 3, 4] give a continuous representative $\phi_n = H \circ \psi$, where ψ is the principal quasiconformal plane homeomorphism solving $\psi_{\bar{w}} = \mu\psi_w$ and H is entire. Since $\psi(w) = w + O(1/w)$ and ϕ_n has the displayed principal expansion, $H(w) = w + O(1/w)$ near infinity. Liouville’s Theorem makes H the identity. Consequently ϕ_n itself is a homeomorphism. Direct differentiation gives the signs and the Jacobian:

$$(\phi_n)_w = \frac{k+1}{2}((v_n)_Y + i(v_n)_X), \quad (\phi_n)_{\bar{w}} = \frac{k-1}{2}((v_n)_Y - i(v_n)_X), \quad J_{\phi_n} = k|\nabla v_n|^2. \quad (4.3)$$

The map $\phi_n \circ h$ is again a model. It is conformal on G , and its new coefficient is $b(h) + c_n$. A homeomorphism carries complementary components to complementary components, preserving points and Jordan disks; uniform continuity on a compact set containing F preserves the null-family property. For an old continued test U , its new continuation is exactly $U \circ \phi_n^{-1}$, with the same prescribed constant on the corresponding component. Quasiconformal composition preserves its global $W^{1,2}(\hat{\mathbb{C}})$ membership and its zero gradient on the transformed complement. To see also the null-set issue in this statement, approximate the continuous U uniformly and strongly in $W^{1,2}$ by smooth sphere functions. The quasiconformal Dirichlet inequality makes their compositions converge in gradient norm, and uniform convergence identifies the limit. The almost-everywhere chain rule then applies. The forward Lusin N property of ϕ_n maps the source-null exceptions to target-null sets, so the resulting gradient vanishes almost everywhere on $\phi_n(F)$. Approximate maximality therefore gives

$$\Re c_n \leq d. \quad (4.4)$$

For D larger than a disk containing F , the area formula and the boundary Laurent expansion give

$$\int_{D_D} J_{\phi_n} dA = \text{area}(\phi_n(D_D)) = \pi D^2 + o(1) \quad (D \rightarrow \infty).$$

Here $D_D = \{w : |w| < D\}$. The Divergence Theorem, or the weak boundary formula on this smooth exterior circle, gives the exact identity

$$\int_{D_D} (v_n)_Y dA = \int_{\partial D_D} v_n \nu_Y ds = \pi D^2 - \pi \Re c_n;$$

all other Laurent terms have zero contribution. Expanding the square, using (4.3), and then letting $D \rightarrow \infty$ yields

$$\int_{\mathbb{C}} |\nabla(v_n - Y)|^2 dA + (n-1) \int_F |\nabla v_n|^2 dA = 2\pi \Re c_n \leq 2\pi d. \quad (4.5)$$

In particular both terms on the left are controlled, with the stated normalization constant.

We next obtain compactness without a uniform quasiconformal distortion bound. A coordinate of a plane homeomorphism attains its maximum and minimum on the boundary of every relatively compact open set: an interior coordinate extremum contradicts openness. For almost every circle centered at x , the Sobolev circle restriction and Cauchy–Schwarz therefore give, for $0 < r < R$,

$$(\operatorname{osc}_{\overline{D(x,r)}} v_n)^2 \log(R/r) \leq 2\pi \int_{D(x,R)} |\nabla v_n|^2 \, dA. \quad (4.6)$$

One obtains this inequality by bounding the oscillation on the smaller disk by that on a circle of radius $t \in (r, R)$, bounding the latter by the integral of $|\nabla v_n|$ around that circle, and integrating with respect to dt/t . Equation (4.5) bounds the right-hand side locally, uniformly in n . Exterior univalence and principal normalization, on a common exterior disk, give

$$|v_n(w) - Y| \leq C/|w| \quad (|w| \geq R_0), \quad (4.7)$$

where C, R_0 depend only on the common bound for F , hence only on B . The maximum and minimum principles give uniform bounds on compact sets as well. Thus (4.6) gives local equicontinuity.

Pass to a locally uniform and locally Sobolev-weak subsequential limit v . Lower semicontinuity proves (4.1), and the second term of (4.5) forces $\nabla v = 0$ almost everywhere on F . The function is therefore constant on the connected interior of each Jordan disk component, and continuity extends the same value to its boundary. Singleton components need no argument. The coordinate maximum and minimum principles and the exterior estimate pass to the limit.

Finally, allow the nearly maximizing models to vary and let $d \rightarrow 0$. The same modulus and exterior bounds give compactness on every compact set. Any subsequential limit of $v - Y$ has zero distributional gradient by (4.1), hence is constant. Estimate (4.7), with arbitrarily large $|w|$, makes the constant zero. This proves the uniform compact-set closeness asserted in the proposition. $\square$

Lemma 4.2. *Let F and v be as in Proposition 4.1. Every point $x \in \mathbb{C}$ lies in a compact nondegenerate continuum of its v -level contained in any prescribed closed ball of positive radius centered at x and meeting that ball's boundary. If $U \in C(\mathbb{C}) \cap W_{\text{loc}}^{1,2}(\mathbb{C})$ is real-valued and $\nabla U = 0$ almost everywhere on F , then*

$$\operatorname{area}((U, v)(F)) = 0, \quad \operatorname{area}((U, X)(F)) = 0. \quad (4.8)$$

Proof. For the approximating homeomorphism ϕ_n , the inverse image of the horizontal line through $\phi_n(x)$ is a proper simple curve through x . Stop one branch at its first exit from the prescribed ball. The resulting compact arcs have a Hausdorff convergent subsequence. Their limit is connected, contains x , and meets the ball boundary. Local uniform convergence $v_n \rightarrow v$ places it in the actual level $v^{-1}(v(x))$. This proves the first assertion pointwise.

We use the scalar Sobolev Coarea Theorem, due to Federer [5], in the precise-representative form of Malý, Swanson, and Ziemer [9, Theorem 1.1]. With domain dimension 2, target dimension 1, and exponent 2, it gives, for every nonnegative Borel weight a supported in a bounded region,

$$\int_{\mathbb{R}} \int_{v^{-1}(t)} a \, d\mathcal{H}^1 \, dt = \int_{\mathbb{C}} a |\nabla v| \, dA. \quad (4.9)$$

Continuity makes v equal everywhere to its precise representative, so these are the full pointwise level sets. In particular, almost every level has finite length in any fixed compact box. Formula (4.9) also implies that planar null sets have zero length on almost every level.

Choose a closed box Q with F in its interior. On a neighborhood of Q , smooth approximation of the continuous Sobolev function U provides $U_j \rightarrow U$ uniformly on Q with

$$\sum_j \|\nabla U_j - \nabla U\|_{L^2(Q)} < \infty.$$

Let g be a nonnegative Borel representative of $|\nabla U|$. By (4.9) and Cauchy–Schwarz, outside one null set of values t we have simultaneously

$$\begin{aligned} \mathcal{H}^1(Q \cap v^{-1}(t)) &< \infty, & \int_{Q \cap v^{-1}(t)} g \, d\mathcal{H}^1 &< \infty, \\ \sum_j \int_{Q \cap v^{-1}(t)} |\nabla U_j - \nabla U| \, d\mathcal{H}^1 &< \infty, & \int_{F \cap v^{-1}(t)} g \, d\mathcal{H}^1 &= 0. \end{aligned}$$

The last assertion uses $g = 0$ almost everywhere on F ; the coarea identity makes the choice on the planar-null exceptional set irrelevant. For every such good level and *every* rectifiable simple arc $\gamma \subset Q \cap v^{-1}(t)$, the smooth arc inequality passes to the limit to give

$$|U(x) - U(y)| \leq \int_{\gamma} g \, d\mathcal{H}^1 \quad (x, y \text{ the endpoints of } \gamma). \quad (4.10)$$

Indeed the gradient error on any arc is bounded by the error on the entire level section, whereas uniform convergence controls both endpoints. Thus the exceptional levels do not depend on the choice of arc or its endpoints.

Fix a good level. Its nondegenerate components in Q form an at most countable family of finite-length continua. Each such continuum is locally connected: failure of local connectedness, by boundary bumping, would give infinitely many disjoint subcontinua of diameter bounded below, contrary to finite length. A locally connected metric continuum is locally arcwise connected, so its connected relatively open subsets are arcwise connected. All their simple arcs are rectifiable, since their lengths are bounded by the length of the whole continuum.

Let K be one of these components and $A = F \cap K$. The finite measure $g\mathcal{H}^1|_K$ vanishes on A . Given $\epsilon > 0$, choose a relatively open neighborhood O of A in K with $\int_O g \, d\mathcal{H}^1 < \epsilon$. Local connectedness and second countability make its connected components O_i a countable family of relatively open sets. Joining any two points of O_i by a simple arc there and using (4.10) shows

$$\text{diam } U(O_i) \leq \int_{O_i} g \, d\mathcal{H}^1.$$

Since each $U(O_i)$ is an interval, the outer length of $U(A)$ is at most ϵ , and hence is zero. Every point of $F \cap v^{-1}(t)$ belongs to a nondegenerate component of $Q \cap v^{-1}(t)$ by the first assertion of the lemma, using a small ball inside Q . Thus the countable family of components covers *every* such point, and $U(F \cap v^{-1}(t))$ has length zero. The compact set $(U, v)(F)$ now has almost every horizontal section of length zero. Fubini proves its area is zero.

For the companion pair, the continuous Sobolev representative has the absolute continuity on lines (ACL) property. Thus U is absolutely continuous on almost every vertical line. Fubini gives $\partial_Y U = 0$ almost everywhere on that line's intersection with F . The one-dimensional area inequality for an absolutely continuous function therefore makes the image of this intersection length-null. Applying Fubini to the compact set $(U, X)(F)$ proves the second equality in (4.8). $\square$

5 Sobolev extension across removable dust

Throughout this section, $B \subset \mathbb{C}$ is compact and totally disconnected, and $G = \widehat{\mathbb{C}} \setminus B$. For a finite vector of scalar functions, the gradient norm means the L^2 norm of its matrix of gradients. Small changes of an old test will always mean small changes both uniformly on the sphere and in gradient norm on G .

Theorem 5.1 (Continuous Sobolev extension). *If $B \subset \mathbb{C}$ is compact, totally disconnected, and conformally removable, then every continuous real function u on $\widehat{\mathbb{C}}$ whose restriction to $G = \widehat{\mathbb{C}} \setminus B$ belongs to $W^{1,2}(G)$ belongs to $W^{1,2}(\widehat{\mathbb{C}})$.*

The next lemma supplies small-energy barriers around the source ends. Its stability assertion will let us retain those barriers while adding further tests.

Lemma 5.2 (Appending separating tests). *There is a radius D_0 , depending only on B , with the following property. Suppose $u_1, \dots, u_m$ satisfy the hypotheses of Proposition 3.2. Given $\delta, \eta > 0$, they may be replaced by tests $\tilde{u}_i$ satisfying*

$$\|\tilde{u}_i - u_i\|_\infty < \delta, \quad \|\nabla(\tilde{u}_i - u_i)\|_{2,G} < \delta,$$

and two tests p, q may be appended, so that the enlarged finite list still satisfies the pair-null condition (3.1). There are finitely many smooth Jordan curves $\gamma_l \subset G \cap D_{D_0}$ whose bounded sides cover B , and smooth functions $\Phi_l : \mathbb{R}^2 \rightarrow [0, 1]$, such that

$$\begin{aligned} \Phi_l(p, q) &= 1 \text{ near } \gamma_l, & \Phi_l(p, q) &= 0 \text{ on a fixed collar of } \partial D_{D_0}, \\ \|\nabla \Phi_l(p, q)\|_{2,G \cap D_{D_0}} &< \eta. \end{aligned} \tag{5.1}$$

The barrier conclusions (5.1), with the same curves and functions Φ_l , survive sufficiently small subsequent changes of the coordinates p, q .

Proof. We construct a continuous pair $P = (p, q)$ and obtain the barriers by composing it with translates of a radial cutoff of small Dirichlet energy. By conformal invariance, composing the cutoff with a model map h does not increase its Dirichlet energy, so we will compare P with $(\Re h, \Im h)$ on a fixed source disk. Proposition 4.1 supplies a height v whose pullback gives the vertical coordinate. We make small range compressions of this pullback and the old tests so that they become constant near every source end corresponding to a nonpoint model hole, then modify the horizontal coordinate there. These modifications preserve pair-nullity and make the gradient error small on the source disk. The vertical error is uniformly small; the horizontal changes are confined to neighborhoods where h has small energy.

Choose $a \geq 1$ with $B \subset D_a$. For every normalized univalent map h on G , the Exterior Area Theorem gives

$$|h(z) - z| \leq a \sqrt{-\log(1 - a^2/|z|^2)} \quad (|z| > a). \tag{5.2}$$

Indeed, apply Cauchy-Schwarz to the Laurent series and the area-theorem coefficient bound. Hence $h(\partial D_{2a}) \subset D_{3a}$, and the omitted set $F = \mathbb{C} \setminus h(G)$ lies on its bounded side. In particular, $F \subset D_{3a}$ for every model of every list. Fix $D_0 = 64a$. The image of $G \cap D_{65a}$ lies in D_{67a} ; these uniform bounds also bound the energy of $(\Re h, \Im h)$ on that source disk.

Choose a smooth radial cutoff Φ , equal to one on a disk of radius $r > 0$ and zero outside D_{4a} , with $\|\nabla \Phi\|_{2,\mathbb{C}} < \eta/3$. Such a cutoff is obtained by smoothing a logarithmic profile and taking r sufficiently small. All its translates have the same derivative bounds. We shall use translates centered at the eventual traces of (p, q) on B .

By Proposition 4.1, take a model h sufficiently close to the supremum of its real Laurent coefficient that the associated height v satisfies

$$\|v - Y\|_{\infty, D_{67a}} \text{ and } \|\nabla(v - Y)\|_{2, \mathbb{C}} \text{ as small as required below,} \quad Y(w) = \Im w. \quad (5.3)$$

Write U_i for the global model continuation of u_i . The function v is constant on every component of F , so $v \circ h$ extends continuously to B by Lemma 2.1.

Flattening the hole values. For a continuous, locally Sobolev real function s of finite Dirichlet energy on a region, the measure

$$\mu_s(I) = \int_{\{s \in I\}} |\nabla s|^2$$

has no atom at any real number: the Sobolev gradient is zero almost everywhere on each fixed level set. Thus, given countably many values, we can surround them by open intervals whose total length and total μ_s -measure are arbitrarily small. If O is their union, then

$$T(t) = \int_0^t \mathbf{1}_{\mathbb{R} \setminus O}(\tau) d\tau$$

is 1-Lipschitz, satisfies $\|T - \text{id}\|_{\infty} \leq |O|$, and is constant near each specified value. The chain rule bounds the squared gradient error by $\mu_s(O)$. This remains valid when the specified values are dense.

Apply this construction to each u_i , at its values at the source ends corresponding to nonpoint components of F , and to v at its values on those components. For v use only the energy on D_{67a} . Denote the flattened old tests by $\tilde{u}_i = T_i \circ u_i$, and put $q_0 = T \circ v \circ h$. Their errors can be arbitrarily small, and the required bounds by δ hold for the old tests. In particular, q_0 is uniformly and energetically close to $\Im h$ on $G \cap D_{65a}$. By Lemma 4.2 and coordinatewise Lipschitz composition, the traces of q_0 form null-area pairs with the flattened old tests. The old pairs also remain null. Each nonpoint-hole end b_j now has a source neighborhood on which all $\tilde{u}_i$ and q_0 are constant.

The horizontal coordinate. Let F_j be the nonpoint component corresponding to b_j , and choose $c_j \in \Re F_j$. Around each b_j choose a ball $B_j \subset D_{2a}$ in the preceding common constancy neighborhood. Its radius may be required to tend to zero and its image in G to lie within $2^{-j}a$ of F_j . The balls can also satisfy

$$\|\mathbf{1}_S \nabla H\|_2 < \tau, \quad S = \bigcup_j B_j, \quad H = (\Re h, \Im h), \quad (5.4)$$

for any prescribed $\tau > 0$: the energy is integrable, and each center is a single source point. Choose logarithmic cutoffs χ_j supported in B_j , equal to one near b_j , with $\sum_j \|\nabla \chi_j\|_2 < \tau$. Put

$$w_j = \chi_j \prod_{i < j} (1 - \chi_i), \quad \rho = \prod_j (1 - \chi_j), \quad p_0 = \rho \Re h + \sum_j w_j c_j \quad \text{on } G. \quad (5.5)$$

These expressions are locally finite in G , since the radii tend to zero and all centers belong to B . They are convex interpolations. On S , all horizontal values used have absolute value at most $4a$. The product rule, applied to the finite formula locally, gives

$$|\nabla p_0 - \rho \nabla \Re h| \leq 8a \sum_j |\nabla \chi_j|, \quad \|\nabla(p_0 - \Re h)\|_{2, G \cap D_{65a}} \leq (1 + 8a)\tau. \quad (5.6)$$

No disjointness of the balls is required.

At b_j , the factor $\chi_j = 1$ makes (5.5) terminate after index j on a neighborhood. It is therefore a finite continuous expression there. At a point end b , write $F_b = \{\zeta_b\}$ and $x_b = \Re \zeta_b$. Early nonzero weights may change the trace from x_b . To prove continuity, compare p_0 with the finite continuous function

$$x_b + \sum_{j \leq N} w_j(z)(c_j - x_b).$$

For sufficiently large N and sufficiently small neighborhoods of b , every later ball meeting that neighborhood has its center near b . Upper semicontinuity of the end correspondence then puts its entire component F_j , and hence c_j , arbitrarily close to ζ_b and x_b . Also $\Re h(z) \rightarrow x_b$ from G . The remaining weight is a convex combination of these close values, so the difference from the displayed finite expression is uniformly arbitrarily small. This proves continuous extension at b , including when early supports overlap. The same estimate controls the prescribed values at nearby ends.

The new pairs are now checked on all of B . On each ball B_j , every coordinate other than p_0 is constant, so a pair involving p_0 has image in a line. Countably many balls contribute zero planar area. Outside S , all ends are point ends and p_0 is their horizontal trace. Apply the horizontal assertion of Lemma 4.2 to $T_i \circ U_i$ and to $T \circ v$, respectively. Their gradients vanish on F , so this proves nullity of the pairs with every old test and with q_0 . Together with the earlier pair checks, this proves the full pair-null condition. Multiply p_0, q_0 by a smooth source cutoff equal to one on D_{65a} and zero outside a larger disk. The resulting p, q are bounded continuous tests of finite energy on G ; no trace or working-disk calculation changes.

Small energy and strict margins. Let $P = (p, q)$ on the working disk. Outside S its first coordinate equals that of H , and write $e = \|q - \Im h\|_\infty$ there. For any translate Φ_l of Φ , with derivative bounds $M_1 = \|D\Phi\|_\infty$ and $M_2 = \|D^2\Phi\|_\infty$, the chain rule gives

$$\|\nabla \Phi_l(P) - \nabla \Phi_l(H)\|_2 \leq M_1 \|\nabla P - \nabla H\|_2 + M_2 e \|\nabla H\|_2 + 2M_1 \|\mathbf{1}_S \nabla H\|_2. \quad (5.7)$$

All norms here are on $G \cap D_{D_0}$. The first and last terms tend to zero by the choices above; the middle term does so by (5.3) and flattening. This estimate does not require a small uniform horizontal error on S . Conformal invariance gives $\|\nabla \Phi_l(H)\|_2 < \eta/3$, so the choices can ensure the strict bound in (5.1), uniformly over all translates.

Take the total vertical uniform error less than a . On B , $|p| \leq 3a$ by convexity and $|q| \leq 4a$, so every trace center has norm at most $5a$. On the collar $63a \leq |z| \leq 65a$, the first coordinate is $\Re h$ and the second differs from $\Im h$ by less than a . Equation (5.2) puts its distance from every trace center above $50a$, well beyond the support radius $4a$. At each $b \in B$, the translate centered at $P(b)$ is one wherever $|P - P(b)| < r$. Choose a smaller source neighborhood whose closure has $|P - P(b)| < r/2$. Compactness and the contour construction in the proof of Lemma 2.1 give finitely many inner contours in such neighborhoods with bounded sides covering B .

Finally, for a fixed saved pair $A = (p, q)$ and any perturbed pair A' ,

$$\|\nabla \Phi_l(A') - \nabla \Phi_l(A)\|_2 \leq M_1 \|\nabla A' - \nabla A\|_2 + M_2 \|A' - A\|_\infty \|\nabla A\|_2. \quad (5.8)$$

The positive plateau and zero-region margins preserve the exact values one and zero under small uniform changes. The positive energy slack and (5.8) preserve the strict norm bound. This proves the stability assertion and the lemma. $\square$

Proof of Theorem 5.1. The empty case is immediate. Fix $0 < \epsilon < 1$ and start with the one-element list u , for which pair-nullity is vacuous. Apply Lemma 5.2 successively with $\eta = 1/k$, saving at stage k its pair, curves, and functions $\Phi_{k,l}$. By the strict margins and (5.8), there is $t_k > 0$ such

that total uniform and gradient changes of that pair below t_k preserve all its saved conclusions with norm less than $1/k$.

At stage n , require each earlier scalar coordinate to change by less than

$$d_n = \frac{2^{-n-3}}{\sqrt{2}} \min(1, \epsilon, t_1, \dots, t_{n-1})$$

in both norms (omit the t 's when $n = 1$). Each stage has only finitely many requirements, and the lemma permits arbitrarily small changes. For every fixed k , the sum of subsequent pair errors is less than $t_k/4$. Thus all saved barriers work at every later finite stage. If a_n is the first coordinate at stage n , then

$$a_n \longrightarrow u_\epsilon \text{ uniformly,} \quad \|u_\epsilon - u\|_\infty < \epsilon, \quad \|\nabla a_n\|_{2,G} \leq \|\nabla u\|_{2,G} + 1. \quad (5.9)$$

Only finite lists are required to be pair-null. In particular, no infinite-list model assertion or limiting pair-null assertion is being used.

Choose a model h_n of each resulting finite list by Proposition 3.2. Lemma 2.2 gives a subsequence converging locally uniformly on G to a normalized univalent map h . We claim that every component of its omitted set is a point. Otherwise, by Lemma 2.1, some source end b has a nondegenerate target component K . One axial projection of K contains an interval J of positive length.

For each k , choose a saved inner contour $\gamma_{k,l}$ enclosing b . Its limiting image strictly encloses K . Uniform convergence on the contour and winding-number stability imply that $h_n(\gamma_{k,l})$ also encloses K for all sufficiently large n in the subsequence. Use the two continued model coordinates at stage n in $\Phi_{k,l}$, on the bounded side of $h_n(\partial D_{D_0})$, and extend by zero outside. Denote this function by $V_{n,k}$. It is continuous and globally Sobolev: the saved zero collar removes any interface jump, and the model coordinates are globally Sobolev. Their gradients vanish on the model holes. The chain rule and conformal invariance therefore give

$$\|\nabla V_{n,k}\|_{2,\mathbb{C}} < 1/k. \quad (5.10)$$

It equals one on $h_n(\gamma_{k,l})$. Moreover all such functions are zero outside one fixed disk D_L , for example with $L = 67D_0/64$ by the uniform bounds in the preceding proof. The radius does not depend on k , n , or the coordinate list.

For definiteness suppose J is the vertical projection interval. Every horizontal line at a height in J meets the inner image contour, where $V_{n,k} = 1$. On almost every such line, the Sobolev restriction is absolutely continuous and agrees with the continuous restriction. It takes value zero at the left and right ends of the segment with horizontal coordinates $-L$ and L . Cauchy–Schwarz, followed by integration over J , gives

$$\int_{\mathbb{C}} |\nabla V_{n,k}|^2 \geq \frac{|J|}{2L} > 0. \quad (5.11)$$

The argument only uses the value one on the contour, as illustrated in Figure 1. Choose k so large that $k^{-2} < |J|/(2L)$, and then choose n sufficiently late for that saved contour. Equations (5.10) and (5.11) contradict each other.

All limit components are therefore singletons. Lemma 2.1 now extends h to an orientation-preserving homeomorphism of the sphere. Removability of B makes it a Möbius map; the normalization $h(z) = z + O(1/z)$ makes it the identity.

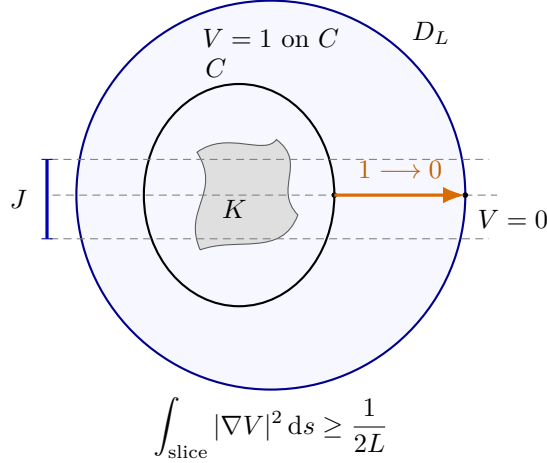


Figure 1: A fixed interval J in a projection of a nondegenerate component K forces a uniform energy cost. Each corresponding line meets the inner contour C , where the barrier is one, and then reaches zero within a segment of length at most $2L$. Integrating over J contradicts arbitrarily small barrier energy. Here $V = V_{n,k}$ and $C = h_n(\gamma_{k,l})$.

Let A_n denote the globally continued first test of the model h_n . Its values on holes are the source endpoint values, and its gradient vanishes there. Consequently

$$\|A_n\|_\infty \leq \|u\|_\infty + 1, \quad \|\nabla A_n\|_{2,\widehat{\mathbb{C}}} = \|\nabla a_n\|_{2,G} \leq \|\nabla u\|_{2,G} + 1.$$

Take a weakly convergent subsequence in $W^{1,2}(\widehat{\mathbb{C}})$. Since $h_n \rightarrow \text{id}$ locally uniformly on G , inverse convergence and (5.9) give $A_n \rightarrow u_\epsilon$ locally uniformly on G . Thus the weak limit equals u_ϵ there. Lemma 2.3 gives $|B| = 0$, so the equality holds almost everywhere on the sphere. This proves global Sobolev membership of u_ϵ , with a bound independent of ϵ . Finally let $\epsilon \downarrow 0$ and use uniform convergence to u and weak Sobolev compactness once more. This proves the theorem. $\square$

6 Null projections of the common point traces

We now apply Theorem 5.1 to a conformal equivalence between circle domains. The scalar functions used below are arranged to be continuous even when a point component corresponds to a disk.

We first record a packing estimate closely related to [11, Lemma 2.1]. We include its short proof from the Hardy–Littlewood Maximal Theorem [16].

Lemma 6.1 (Ball synthesis). *Let $B_j \subset \mathbb{R}^2$ be balls of radii $d_j > 0$. Suppose there are pairwise disjoint measurable sets $E_j \subset B_j$ with $|E_j| \geq c d_j^2$, where $c > 0$ is fixed. For nonnegative $(a_j) \in \ell^2$,*

$$\left\| \sum_j \frac{a_j}{d_j} \mathbf{1}_{B_j} \right\|_{L^2(\mathbb{R}^2)} \leq C_c \left(\sum_j a_j^2 \right)^{1/2}.$$

Proof. Let $g \geq 0$ belong to L^2 , and let Mg be its centered maximal function. For every $x \in B_j$,

the average of g on B_j is at most $4Mg(x)$, since $B_j \subset B(x, 2d_j)$. Consequently

$$\begin{aligned} \int g \sum_j \frac{a_j}{d_j} \mathbf{1}_{B_j} &\leq 4\pi \sum_j a_j d_j \inf_{E_j} Mg \\ &\leq C_c \left(\sum_j a_j^2 \right)^{1/2} \left(\sum_j \int_{E_j} (Mg)^2 \right)^{1/2} \leq C_c \| (a_j) \|_{\ell^2} \|g\|_2. \end{aligned}$$

Here the last inequality uses disjointness and the L^2 Maximal Theorem. Duality proves the assertion for finite sums, and monotone convergence proves it for countable sums. $\square$

For $e \in \{1, i\}$ write $\pi_e(w) = \Re(\bar{e}w)$.

Proposition 6.2 (Common point traces). *Let Ω, Ω' be circle domains containing infinity, and let $f : \Omega \rightarrow \Omega'$ be a conformal equivalence with $f(\infty) = \infty$. Suppose $\partial\Omega$ is conformally removable. Let C be the set of point components of $\widehat{\mathbb{C}} \setminus \Omega$ whose corresponding components in $\widehat{\mathbb{C}} \setminus \Omega'$ are also points, and denote their positions by $f_*(x)$, $x \in C$. For each $e \in \{1, i\}$ and almost every line ℓ parallel to e ,*

$$|\{\pi_e(f_*(x)) : x \in C \cap \ell\}| = 0.$$

Here and below $|\cdot|$ for a subset of a line denotes Lebesgue length.

Proof. We fix e throughout. All complementary components on either side lie in bounded planar regions. Index those pairs (K_j, K'_j) for which at least one component is a disk of positive radius. The index set is finite or countable, since disjoint positive-radius disks have disjoint nonempty interiors. Write

$$K_j = \overline{B}(a_j, r_j), \quad K'_j = \overline{B}(a'_j, s_j),$$

allowing radius zero, and set $\beta_j = \pi_e(a'_j)$. Disjointness and boundedness give

$$\sum_j r_j^2 < \infty, \quad \sum_j s_j^2 < \infty. \tag{6.1}$$

The other components are precisely the common points in C .

Scalar data and shrinking neighborhoods. Choose a smooth planar cutoff χ , equal to one on a sufficiently large disk containing the source complement and all the neighborhoods introduced below, with its transition region compactly contained in Ω . On Ω put

$$b = \chi \pi_e(f),$$

and extend b by β_j on K_j , by $\pi_e(f_*(x))$ on C , and by zero at infinity. This is bounded. Indeed, $f(\infty) = \infty$ implies that the image of a bounded source region is bounded. The area formula for the univalent map f then shows that b has finite Dirichlet energy on Ω ; the cutoff-derivative term is supported in a compact subset of Ω .

Although b need not be continuous at K_j , it is continuous at every common point. Moreover, the cluster values near K_j lie in $[\beta_j - s_j, \beta_j + s_j]$. These statements include approaches through other complementary components and follow from the upper-semicontinuous correspondence in Lemma 2.1.

Choose $\varepsilon_j > 0$ tending to zero with $\sum_j \varepsilon_j^2 < \infty$, and put

$$I_j = (\beta_j - s_j - \varepsilon_j, \beta_j + s_j + \varepsilon_j), \quad l_j = |I_j|.$$

Thus $(l_j) \in \ell^2$. Fix A so large that all these intervals and all values of b lie in $(-A, A)$. Choose positive Δ_j with

$$\sum_{i>j} \Delta_i^2 < \frac{1}{2} \Delta_j^2; \quad (6.2)$$

for example, a sufficiently small multiple of 2^{-j} suffices. For each integer $m \geq 1$, define the balloons

$$B_{j,m} = B(a_j, d_{j,m}), \quad d_{j,m} = \begin{cases} (1 + 1/m)r_j, & r_j > 0, \\ \Delta_j/m, & r_j = 0. \end{cases}$$

These balls are all in one bounded region, and decrease to K_j as $m \rightarrow \infty$. Choose radial Lipschitz functions $0 \leq \lambda_{j,m} \leq 1$ that vanish on an open neighborhood of K_j , equal one outside $B_{j,m}$, and satisfy

$$|\nabla \lambda_{j,m}| \leq \frac{C_m}{d_{j,m}} \mathbf{1}_{B_{j,m}} \quad \text{almost everywhere.} \quad (6.3)$$

One may take $C_m = 2(m+1)$, interpolating radially between $(1 + 1/(2m))r_j$ and $(1 + 1/m)r_j$ when $r_j > 0$, and between $d_{j,m}/2$ and $d_{j,m}$ otherwise.

For the moment fix m and suppress it from the notation. We will transform the scalar value $b(z)$ before applying Sobolev extension. Near K_j the transformation should be constant on I_j , so it does not distinguish the possible cluster values of b . The cutoff λ_j imposes no restriction outside B_j . Taking the infimum of the cutoffs whose intervals contain a given scalar value enforces these requirements simultaneously, including where the intervals and neighborhoods overlap. Integrating in the scalar variable makes the spatial cost proportional to l_j , as the estimate below will show. Define

$$L(z, t) = \inf_{j: t \in I_j} \lambda_j(z), \quad F(z, t) = \int_{-A}^t L(z, s) \, ds, \quad U(z) = F(z, b(z)), \quad (6.4)$$

with the empty infimum equal to one. We shall prove that U is continuous and belongs to $W^{1,2}(\widehat{\mathbb{C}})$. *An energy density.* Set

$$P = C_m \sum_j \frac{l_j}{d_j} \mathbf{1}_{B_j}. \quad (6.5)$$

This belongs to $L^2(\mathbb{C})$. For the positive-radius sublist, the interiors of the original disks K_j are disjoint subsets of B_j of area at least $\pi d_j^2/4$, so Lemma 6.1 applies. For the zero-radius sublist use instead

$$E_j = B_j \setminus \bigcup_{i>j, r_i=0} B_i.$$

These sets are pairwise disjoint: if $i < j$, then E_i excludes $B_j \supset E_j$. Since all zero-radius balloons in this system use the same factor $1/m$, (6.2) gives

$$|E_j| \geq \pi d_j^2 - \pi \sum_{i>j, r_i=0} d_i^2 \geq \frac{1}{2} \pi d_j^2.$$

Apply Lemma 6.1 separately to this sublist and then add the two L^2 bounds. Overlap between the two sublists is irrelevant.

Continuity of the masks. We spell out the accumulation issue. For fixed m , $d_j \rightarrow 0$. If distinct balloon supports accumulate at a common point x , their centers and source components approach x . By Lemma 2.1 their target components approach $\{f_*(x)\}$; as $\varepsilon_j \rightarrow 0$, the corresponding intervals I_j approach the scalar $b(x)$. Consequently, given $\delta > 0$, there are a neighborhood V of x and a

finite initial list such that every remaining balloon meeting V has its interval in $(b(x) - \delta, b(x) + \delta)$. The finite-mask version of F differs from F by at most 2δ on V , uniformly in the upper endpoint. Since the finite-mask version is continuous in its two variables and b is continuous at x , U is continuous there.

For an indexed component K_j , choose a neighborhood V_j of the entire closed component on which $\lambda_j = 0$ and $b \in I_j$. This is possible because I_j has a positive margin around all its cluster values. On V_j the upper endpoint in (6.4) may be replaced by the fixed value β_j . All sufficiently late intervals of balloons meeting a sufficiently small neighborhood of K_j lie in I_j . To verify the uniformity of this assertion, a failure would give $i_\nu \rightarrow \infty$ and $z_\nu \in B_{i_\nu}$ with $\text{dist}(z_\nu, K_j) \rightarrow 0$, but $I_{i_\nu} \not\subset I_j$. Passing to a subsequence gives $z_\nu \rightarrow x \in K_j$; the shrinking radii imply $K_{i_\nu} \rightarrow \{x\}$. Component correspondence puts their target components arbitrarily close to K'_j , and the vanishing extra margins then force $I_{i_\nu} \subset I_j$, a contradiction. These late intervals have no effect on L near K_j : there $\lambda_j = 0$, so $L(z, t) = 0$ for $t \in I_j$. Thus near K_j the fixed-endpoint expression is a finite continuous mask. This proves continuity there, including when $r_j = 0 < s_j$.

In Ω the cutoffs are locally finite, and near infinity they are all one. Hence the remaining continuity assertions are immediate, with $U = A$ near infinity.

Sobolev regularity and the residual dust. Let L_N, F_N, U_N denote the masks using just the first N indices. For $t \in [-A, A]$ and any x, y ,

$$|F_N(x, t) - F_N(y, t)| \leq \sum_{j \leq N} l_j |\lambda_j(x) - \lambda_j(y)|. \quad (6.6)$$

This follows from the elementary difference bound for finite minima, integrated over the intervals on which a cutoff can occur. Changing the upper endpoint costs at most its change, since $0 \leq L_N \leq 1$. It follows that U_N has gradient bounded by $|\nabla b| + P$ locally in Ω . On V_j , once $N \geq j$, its endpoint can be fixed at β_j , and its gradient is bounded by P .

Bounded pointwise convergence of the finite masks gives local strong L^2 convergence. Weak Sobolev compactness therefore proves that U is locally Sobolev on

$$O = \Omega \cup \bigcup_j V_j,$$

with the same gradient bounds. These yield the finite total energy estimate

$$\int_O |\nabla U|^2 \leq 2 \int_\Omega |\nabla b|^2 + 2 \int_C P^2 < \infty. \quad (6.7)$$

Here the open union is integrated once; no multiplicity is introduced by overlapping neighborhoods.

The residual set $B_* = \widehat{\mathbb{C}} \setminus O$ is compact and bounded, and consists only of common point components. Every connected subset of B_* lies in one component of $\widehat{\mathbb{C}} \setminus \Omega$, hence is a singleton. Thus B_* is totally disconnected. Moreover $B_* \subset \partial\Omega$, and so it is conformally removable by inclusion. Theorem 5.1, applied to the continuous bounded function U and (6.7), proves $U \in W^{1,2}(\widehat{\mathbb{C}})$. If B_* is empty, this conclusion already follows from the local Sobolev statement and the energy bound. We have established the conclusion for every mask system m ; denote its function and density by U_m, P_m .

A typical line and compression. Choose a line ℓ parallel to e on which all U_m are locally absolutely continuous and all P_m are locally integrable. Almost every such line has these properties, since there are only countably many systems. We also require

$$|\partial\Omega \cap \ell| = 0, \quad \sum_{j: B_{j,1} \cap \ell \neq \emptyset} l_j < \infty. \quad (6.8)$$

The first condition holds on almost every line by Lemma 2.3 and Fubini. For the second, write ℓ_τ for the parallel line with transverse coordinate τ . Tonelli gives

$$\int_{\mathbb{R}} \sum_{j: B_{j,1} \cap \ell_\tau \neq \emptyset} l_j d\tau = 2 \sum_j l_j d_{j,1} < \infty,$$

by Cauchy–Schwarz and (6.1)–(6.2). Fix a line satisfying all these conditions.

For $N \geq 1$ let

$$I^{(N)} = \bigcup_{\substack{j > N \\ B_{j,1} \cap \ell \neq \emptyset}} I_j, \quad H_N(t) = \int_{-A}^t \mathbf{1}_{\mathbb{R} \setminus I^{(N)}}(s) ds.$$

Equation (6.8) implies $|I^{(N)}| \rightarrow 0$. For any $x \in C \cap \ell$, the distance from x to each of the first N closed components is positive. Some system m therefore has an open line neighborhood J of x missing its first N balloons. On J , every active cutoff belongs to the tail appearing in $I^{(N)}$, and hence

$$L_m(z, t) \geq \mathbf{1}_{\mathbb{R} \setminus I^{(N)}}(t).$$

The fixed-endpoint variation estimate extends to the infinite mask: for any segment $[x, y] \subset \ell$ and fixed $t \in [-A, A]$,

$$|F_m(x, t) - F_m(y, t)| \leq \int_{[x, y]} P_m ds. \quad (6.9)$$

Indeed (6.3) and (6.6) give this for finite masks, and dominated convergence in the integration variable gives the limit. The estimate uses only the globally Lipschitz cutoffs, so it remains valid when the segment crosses indexed components where b is discontinuous.

If $x, y \in C \cap J$, compare the two upper endpoints at one fixed location, and then use (6.9). The nonnegative-integrand inequality above gives

$$\begin{aligned} |H_N(b(x)) - H_N(b(y))| &\leq |U_m(x) - U_m(y)| + \int_{[x, y]} P_m ds \\ &\leq \int_{[x, y]} (|U'_m| + P_m) ds. \end{aligned} \quad (6.10)$$

For example, when $b(x) \leq b(y)$, first bound the compressed increment by $F_m(x, b(y)) - F_m(x, b(x))$ and then replace $F_m(x, b(y))$ by $F_m(y, b(y))$. The other order is identical.

The set $C \cap J$ has length zero. A general elementary consequence of (6.10) is therefore

$$|H_N(b(C \cap J))| = 0.$$

To see this with outer length, work first in a compact subinterval of J . Cover its null set of common points by an open set on which the integral of $|U'_m| + P_m$ is arbitrarily small. On each component interval, (6.10) bounds the diameter of the image of the common points by that integral. Summing gives arbitrarily small interval covers of the image. A countable exhaustion proves the assertion on J .

We recover the uncompressed values by the one-dimensional area formula. The common-point set is Borel, since it is the compact complement with the countable union of the indexed closed components removed. Its trace map is continuous in the relative topology. Thus $b(C \cap J)$ is analytic and Lebesgue measurable. Put $S = b(C \cap J) \setminus I^{(N)}$. The nondecreasing Lipschitz function H_N has derivative one almost everywhere outside $I^{(N)}$, and hence

$$|S| = \int_S |H'_N| = \int_{\mathbb{R}} \#\{t \in S : H_N(t) = u\} du = 0.$$

The last integrand is supported on the null set $H_N(b(C \cap J))$. This reasoning allows both noninjectivity of H_N and its null set of derivative exceptions.

For fixed N , countably many of the neighborhoods J suffice to cover $C \cap \ell$, by the countable base of the line. It follows that $b(C \cap \ell) \setminus I^{(N)}$ is null. Finally $|I^{(N)}| \rightarrow 0$, so $|b(C \cap \ell)| = 0$. Since $b(x) = \pi_e(f_*(x))$ on common points, this is the required conclusion. Empty or finite indexed lists are included by interpreting the corresponding sums and tails in the evident way. $\square$

7 The length–area argument

The trace statement now supplies the line estimates needed for a flux lower bound. Comparing that bound with the source and target area bounds will force the normalized conformal map to have derivative of unit modulus and corresponding disks to have equal radii.

Proof of Theorem 1.1. If $\Omega = \widehat{\mathbb{C}}$, its conformal image is also the sphere, by compactness, and its automorphisms are Möbius transformations. Assume henceforth that the complement is nonempty. Apply separate Möbius changes in source and target to move a domain point and its image to infinity. A further target affine change gives

$$f(z) = z + \sum_{n \geq 1} \alpha_n z^{-n} \quad \text{near infinity.} \quad (7.1)$$

These changes preserve circle domains and conformal removability. All complementary components are now bounded Euclidean disks or points. Their correspondence is supplied by Lemma 2.1.

Index by j precisely those corresponding pairs for which at least one component is nondegenerate. Write

$$K_j = \overline{D}(a_j, r_j), \quad K'_j = \overline{D}(b_j, s_j),$$

allowing a zero radius to mean a singleton. The index set is finite or countable, and

$$\sum_j r_j^2 < \infty, \quad \sum_j s_j^2 < \infty, \quad \sum_j r_j s_j < \infty. \quad (7.2)$$

Indeed the positive-radius disks on each side have disjoint interiors in a bounded region, and the last assertion follows by Cauchy–Schwarz. All other complementary components are common points: both the source and target component are singletons. Let τ denote their correspondence.

Choose R strictly larger than a radius containing the source complement. Define a vector field F on $\overline{D}_R$ by using f in Ω , the center b_j on K_j , and $\tau(p)$ at every common point p . This is a bounded measurable field. Boundedness follows from Lemma 2.1, or from the bounded side of the image outer contour. Measurability follows since there are countably many indexed closed components, the common-point set is Borel, and its singleton trace map is continuous relative to that set. No continuity of F at an indexed component is asserted.

A slice estimate. Remove disjoint slightly enlarged concentric disks around the first N indexed source components, using a positive enlargement also when $r_j = 0$. Denote the remaining round region by A_N . The sizes of the enlargements will tend to zero with N fixed.

Consider a closed component segment $J = [x, y]$ of an axial slice of A_N , directed toward increasing coordinate. At domain points use the projected value of f ; at a complementary component use the entire projection interval of its corresponding target component. These fibers are compact intervals in a fixed bounded interval, and their relation has closed graph by Lemma 2.1.

Their union over J is connected. To verify this directly, a missing value t between two attained values would place each fiber wholly below or wholly above t . Upper semicontinuity makes these two classes open in J , giving a separation of the segment.

Consequently all values between the selected endpoint values are covered by the projected images of domain arcs, common-point traces, and the full intervals of met indexed components. By Proposition 6.2, the common traces have length zero on almost every such line. The countably many domain arcs have total projected-image length at most the integral of $|f'|$ over them. Each met indexed component contributes at most $2s_j$. Writing F_{ax} for the coordinate along the slice, we obtain

$$F_{\text{ax}}(y) - F_{\text{ax}}(x) \leq \int_{J \cap \Omega} |f'| \, ds + \sum_{\substack{j > N \\ K_j \cap J \neq \emptyset}} 2s_j \quad (7.3)$$

for almost every line in either direction. In fact the same argument bounds the absolute endpoint difference. Endpoints on an enlarged circle use the same field F for both slicing directions.

Multiplicity of the tail terms. For fixed N the enlargements can be chosen so that each remaining positive-radius source component meets at most one enlarged disk. For $N \geq 2$, let $\delta > 0$ be the minimum separation of the first N source components. Make their enlargements small enough to retain gaps greater than $\delta/2$. A remaining disk of diameter less than $\delta/2$ cannot meet two of them. Only finitely many remaining disks have larger diameter, and their positive distances from the first N components permit further shrinking of the enlargements so that these finitely many disks meet none. This argument includes point components among the first N . For $N = 1$ there is nothing to prove, and for $N = 0$ nothing is removed.

A line meets a remaining disk in an interval; deleting at most one enlarged-disk interval leaves at most two retained pieces. Thus its cost in (7.3) is counted at most twice. Integrating transversely gives at most $8r_js_j$ per direction and at most $16r_js_j$ for both directions. Indexed source point components are missed by almost every axial line, even when their target components are disks. All exceptional indexed point lines form a countable set. The tail estimates are integrable by (7.2).

Flux. Integrating (7.3) over horizontal and vertical lines gives the flux through the outer circle minus the flux through the enlarged inner circles. This is the elementary projection formula for arclength on a circle: the transverse projection Jacobians are $|n_X|$ for horizontal slices and $|n_Y|$ for vertical slices. It applies to bounded measurable boundary values. The normal on an inner circle is taken outward from its removed disk. The two domain integrals are bounded by $2 \int_{\Omega \cap D_R} |f'| \, dA$.

On the outer circle $F = f$, and (7.1) gives exactly

$$\int_{\partial D_R} F \cdot n \, ds = \operatorname{Re} \int_0^{2\pi} f(Re^{it}) e^{-it} R \, dt = 2\pi R^2. \quad (7.4)$$

On the enlarged circle about K_j , subtract the constant target center b_j , whose flux is zero. As its radius decreases to r_j , Lemma 2.1 puts all its assigned values within distance $s_j + o(1)$ of b_j , uniformly along the circle. Hence its inner outward flux I_j satisfies

$$\limsup I_j \leq 2\pi r_js_j. \quad (7.5)$$

When $r_j = 0$ the circle length tends to zero; when $s_j = 0$ the target displacement tends to zero. Thus both types of mismatch are covered.

Moving the inner fluxes to the right and letting the finitely many enlargements shrink, while retaining the preceding multiplicity property, gives

$$2\pi R^2 \leq 2 \int_{\Omega \cap D_R} |f'| \, dA + 2\pi \sum_{j \leq N} r_j s_j + 16 \sum_{j > N} r_j s_j.$$

Let N exhaust the indexed components. In the finite or empty case the corresponding terminal sum is already zero. We conclude that

$$2\pi R^2 \leq 2 \int_{\Omega \cap D_R} |f'| \, dA + 2\pi \sum_j r_j s_j. \quad (7.6)$$

Area and equality. The source domain portion and the interiors of its complementary disks are disjoint subsets of D_R , so

$$\int_{\Omega \cap D_R} 1 \, dA + \pi \sum_j r_j^2 \leq \pi R^2. \quad (7.7)$$

The image of ∂D_R is a Jordan contour. Its bounded side contains $f(\Omega \cap D_R)$ and all target complementary components: the compact exterior source disk is mapped homeomorphically onto the closed Jordan side containing infinity. Indeed its interior is mapped openly, its boundary is the image contour, and compactness makes its image closed in that side; connectedness then identifies the entire side. These pieces are disjoint. By the Laurent area formula its enclosed area is

$$\pi R^2 - \pi \sum_{n \geq 1} n |\alpha_n|^2 R^{-2n} \leq \pi R^2.$$

Conformal change of variables therefore gives

$$\int_{\Omega \cap D_R} |f'|^2 \, dA + \pi \sum_j s_j^2 \leq \pi R^2. \quad (7.8)$$

No area-zero assertion about the target dust is needed: any area in its uncountably many singleton components is simply omitted from the left side of this inequality.

Adding (7.7) and (7.8), then using (7.6), yields

$$\boxed{\int_{\Omega \cap D_R} (|f'| - 1)^2 \, dA + \pi \sum_j (r_j - s_j)^2 \leq 0.} \quad (7.9)$$

Both terms are nonnegative. Continuity of the integrand shows that $|f'| = 1$ throughout the nonempty open set $\Omega \cap D_R$. On any disk compactly contained in this set, the Open Mapping Theorem makes f' constant, so $f(z) = \lambda z + c$ there. The meromorphic Identity Theorem on the connected domain Ω gives this equality everywhere in Ω . Finally (7.1) forces $\lambda = 1$ and $c = 0$. Undoing the source and target changes of coordinates proves that the original equivalence is the restriction of a Möbius transformation. $\square$

References

- [1] Lars Ahlfors and Lipman Bers. Riemann's mapping theorem for variable metrics. *Ann. of Math. (2)*, 72(2):385–404, 1960. doi:10.2307/1970141.

- [2] Lars V. Ahlfors. *Conformal Invariants: Topics in Geometric Function Theory*. McGraw-Hill Series in Higher Mathematics. McGraw-Hill, New York, 1973. Reprinted by the American Mathematical Society, 2010. [Publisher record](#).
- [3] Lars V. Ahlfors. *Lectures on Quasiconformal Mappings*, volume 38 of *University Lecture Series*. American Mathematical Society, Providence, RI, second edition, 2006. [doi:10.1090/ulect/038](#).
- [4] Kari Astala, Tadeusz Iwaniec, and Gaven Martin. *Elliptic Partial Differential Equations and Quasiconformal Mappings in the Plane*, volume 48 of *Princeton Mathematical Series*. Princeton University Press, Princeton, NJ, 2009. [Publisher contents](#).
- [5] Herbert Federer. *Geometric Measure Theory*, volume 153 of *Grundlehren der mathematischen Wissenschaften*. Springer-Verlag, 1969. Reprinted in *Classics in Mathematics*, 1996. [doi:10.1007/978-3-642-62010-2](#).
- [6] Zheng-Xu He and Oded Schramm. Fixed points, Koebe uniformization and circle packings. *Ann. of Math. (2)*, 137(2):369–406, 1993. [doi:10.2307/2946541](#).
- [7] Zheng-Xu He and Oded Schramm. Rigidity of circle domains whose boundary has σ -finite linear measure. *Invent. Math.*, 115:297–310, 1994. [doi:10.1007/BF01231761](#).
- [8] Peter W. Jones and Stanislav K. Smirnov. Removability theorems for Sobolev functions and quasiconformal maps. *Ark. Mat.*, 38(2):263–279, 2000. [doi:10.1007/BF02384320](#).
- [9] Jan Malý, David Swanson, and William P. Ziemer. The co-area formula for Sobolev mappings. *Trans. Amer. Math. Soc.*, 355(2):477–492, 2003. [doi:10.1090/S0002-9947-02-03091-X](#). [arXiv:math/0112008v1](#).
- [10] R. L. Moore. Concerning upper semi-continuous collections of continua. *Trans. Amer. Math. Soc.*, 27(4):416–428, 1925. [doi:10.1090/S0002-9947-1925-1501320-8](#).
- [11] Dimitrios Ntalampekos. Rigidity and continuous extension for conformal maps of circle domains. *Trans. Amer. Math. Soc.*, 376(7):5221–5239, 2023. [doi:10.1090/tran/8923](#).
- [12] Dimitrios Ntalampekos. CNED sets: countably negligible for extremal distances. *Selecta Math. (N.S.)*, 30, 2024. Paper No. 61. [doi:10.1007/s00029-024-00951-5](#).
- [13] Dimitrios Ntalampekos and Malik Younsi. Rigidity theorems for circle domains. *Invent. Math.*, 220(1):129–183, 2020. [doi:10.1007/s00222-019-00921-1](#).
- [14] OpenAI. Koebe’s Circle-Domain Conjecture. OpenAI Math Release preprint [OAI:Koebes-Circle-Domain-Conjecture-September-23-2026](#), 2026.
- [15] Kai Rajala. Rigid circle domains with non-removable boundaries. *Proc. Lond. Math. Soc. (3)*, 131(3):e70081, 2025. 18 pp. [doi:10.1112/plms.70081](#).
- [16] Elias M. Stein. *Singular Integrals and Differentiability Properties of Functions*, volume 30 of *Princeton Mathematical Series*. Princeton University Press, Princeton, NJ, 1970.
- [17] Malik Younsi. Removability, rigidity of circle domains and Koebe’s conjecture. *Adv. Math.*, 303:1300–1318, 2016. [doi:10.1016/j.aim.2016.08.039](#).